
Hidden Votes, Open Stage

Summary

In Dancing with the Stars (DWTS), outcomes depend on both judges' scores and fan votes. While judges focus on technical performance, fan votes are largely popularity-driven, often allowing low-scoring contestants to advance. These controversies raise concerns about fairness and professionalism, motivating a quantitative framework to infer latent fan voting behavior and assess different elimination systems.

For Task 1, we tackle the inverse problem of unobservable fan votes using a two-stage latent-variable framework. Fan support is represented by an inferred vote share driven mainly by intrinsic popularity and judges' scores. An **initial Bayesian model estimated via MCMC** captures voting randomness but fails on extreme high-popularity, low-score cases. We thus propose a **constrained optimization model with Total Variation regularization**, casting fan popularity inference as a **Bayesian Maximum A Posterior (MAP)** under strict elimination constraints. Assuming smooth temporal evolution of popularity (a component of fan votes), the model enforces near-complete **consistency with historical eliminations**, achieving **99.14% accuracy**. A **certainty graphical analysis** further quantifies the interval over which estimates maintain a 95% elimination probability.

For Task 2, using the inferred fan votes, we systematically compare rank-based and percentage-based vote combination methods across 34 seasons. By quantifying inconsistency rates and measuring alignment with judges' preferences and fan preferences, our results show that **the percentage-based system** is more inclined to reflect fan vote preferences. However, in controversial scenarios with strong judge–fan disagreement, neither the rank-based nor the percentage-based system consistently produces eliminations aligned with professional evaluation. By contrast, **an additional system** in which judges determine the final elimination among the bottom two contestants effectively accelerates the elimination of highly popular but low-performing contestants. From the perspective of competition professionalism, *we therefore recommend a hybrid system combining rank-based system with a final judges' decision.*

For Task 3, we employ a **Linear Mixed Effects Model (LMM)** to examine the influence of professional dance partners and contestant characteristics on competition outcomes. Our analysis indicates that these factors affect both judges' scores and fan votes, but the magnitude and direction of their effects differ between the two, highlighting **heterogeneous impacts** across evaluation sources.

For Task 4, based on previous findings, we propose an improved **dynamic elimination framework** that adjusts the relative weight of judges' scores and fan votes according to performance levels. This design reduces the influence of fan popularity when judges' scores are exceptionally low, thereby mitigating extreme controversies while preserving audience participation.

Finally, a stability and uncertainty analysis is conducted to assess the robustness of the proposed framework, confirming its feasibility. A memo summarizing our key findings and recommendations was also prepared, providing practical guidance for future modifications of elimination rules.

Keywords: DWTS Fan Votes Judge Scores Bayesian MAP LMM Dynamic Weighting

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1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) is a competitive reality television show where celebrity contestants are paired with professional dancers. Each week, contestants perform dances that are scored by a panel of judges and by fan votes. The combined scores determine weekly eliminations and the season's final rankings.

Judges' scores aim to reflect technical skill and performance quality, but they inherently include subjective judgment. Fan votes are influenced not only by dance performance but also by the celebrity's popularity, charisma, and other non-technical factors.

Throughout the show's history, different methods have been used to combine judges' scores and fan votes. In the first two seasons, ranks were summed to produce a combined score. From season 3 to roughly season 27, percentages of total scores were used instead. Around season 28, an additional system was introduced: the bottom two contestants were determined by combined scores, then judges chose who to eliminate.

These variations and the secrecy of fan vote totals have occasionally caused controversy, as some contestants consistently received low judges' scores but advanced far or even won. This motivates the core problem: estimating fan votes and analyzing how different vote-combining methods and contestant characteristics affect outcomes.

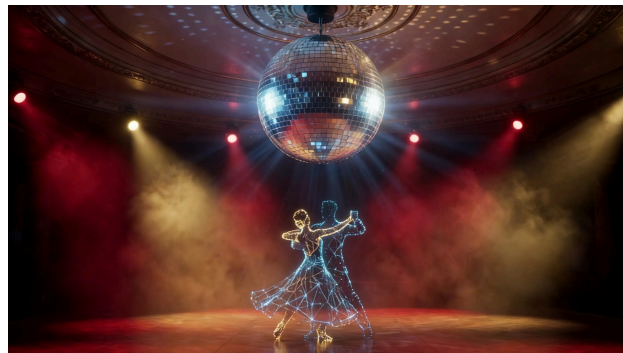


Figure 1: Problem C: Data With The Stars

1.2 Restatement of the Problem

The main objective of this study is to analyze and model the interaction between judges' scores and fan votes in a competitive reality show, in order to identify fairer evaluation methods. Specifically, the problem can be divided into five interrelated questions:

1. **Estimating fan votes:** Develop a mathematical model to estimate weekly fan votes for each contestant. Assess whether the estimates are consistent with the elimination results and quantify the uncertainty of these estimates for different contestants and weeks.
2. **Analyzing vote combination methods:** Compare and evaluate the two approaches used by the show to combine judges' scores and fan votes (rank-based and percentage-based). Examine cases where judges and fans disagree, and investigate how alternative elimination

rules (e.g., judges choosing among bottom contestants) would affect results. Determine which combination of the three methods is preferable based on fairness and consistency.

3. **Impact of professional dancers and contestant characteristics:** Develop models to analyze how the characteristics of celebrities (age, industry, etc.) and professional dancers affect contestants' performance. Investigate whether these characteristics impact judges' scores and fan votes in similar ways.
4. **Proposing an improved system:** Suggest an alternative weekly scoring system that combines fan votes and judges' scores more fairly or in a way that enhances audience engagement, and justify why this system could be adopted by show producers.
5. **Reporting results:** Produce a report of no more than 25 pages detailing findings, and include a one- to two-page memo summarizing the analysis and providing recommendations for future seasons regarding how judge and fan votes should be combined.

1.3 Our work

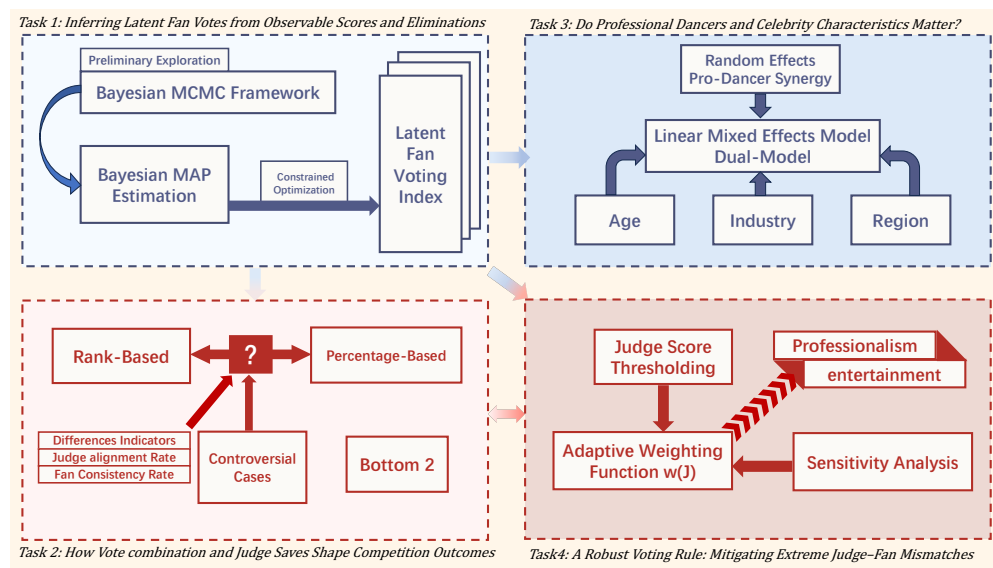


Figure 2: Our Work

2 Preparation of the Models

2.1 Assumptions and Justifications

To simplify our modeling and render the problem solvable, we make the following assumptions:

- **Assumption 1:** Smooth Evolution of Latent Popularity.

Justification: We assume a contestant's latent popularity is inherently stable, changing incrementally over time rather than exhibiting erratic fluctuations between weeks.

- **Assumption 2:** Linearity of Fan Support Factors.
Justification: The unobservable fan vote share is driven primarily by a linear combination of static inherent popularity (Star Power) and dynamic weekly performance (Skill).
- **Assumption 3:** Rational Consistency in Judge Decisions.
Justification: When simulating the “Judge Save” mechanism, we assume judges’ binary voting choices align strictly with their original scoring rankings for that week.

2.2 Notations

Symbol	Definition	Symbol	Definition
i, t	Indices for contestants and weeks	D	Elimination inconsistency rate
$J_{i,t}$	Total judges’ score	C_J, C_U	Judge/Fan alignment rates
S_t	Set of competing contestants	$\hat{J}_{i,t}, \hat{A}_{i,t}$	Standardized scores and vote index
$A_{i,t}$	Latent fan voting index	u_{pro}	Professional dancer random effect
P_i	Static inherent popularity	$w(J)$	Dynamic judge weight
$Q_{i,t}$	Dynamic performance signal	θ	Score threshold parameter
λ	Performance weight parameter	R_{drop}	Ranking decline metric
$U_{i,t}$	Inferred fan vote share	M_{rev}	Vote reversal probability
$S_{i,t}$	Composite elimination score		

3 Task 1: Inferring Latent Fan Votes from Observable Scores and Eliminations

The primary challenge of this task lies in the fact that fan vote counts are not directly observable, whereas the weekly elimination outcomes—jointly determined by judges’ scores and fan votes—are public knowledge. This constitutes a typical **inverse problem** involving latent variables.

To bridge the gap between unobservable votes and observed outcomes, we developed two successive modeling frameworks. We first explored a **Bayesian Stochastic Simulation (Model I)** [1] to capture the probabilistic nature of voting. However, identifying critical limitations in this approach, particularly its failure to explain “outlier” cases like Bobby Bones—led us to develop a more robust **Constrained Optimization & Total Variation (TV) Regularization [2] framework (Model II)**. This evolutionary process ensures that our final estimates are both logically consistent with historical data and mathematically rigorous.

3.1 Model I: Preliminary Exploration - Bayesian MCMC Framework

Fan Vote Generation Logic:

We model fan voting behavior as being driven by a latent **Fan Voting Index** $A_{i,t}$, which is assumed to be positively correlated with the actual number of fan votes received by contestant i in week t . This index represents the underlying propensity of a contestant to attract fan support and

is modeled as a linear combination of inherent popularity and weekly performance:

$$A_{i,t} = (1 - \mu)P_i + \mu Q_{i,t} + \epsilon_{i,t}, \quad \epsilon_{i,t} \sim \mathcal{N}(0, \sigma^2). \quad (1)$$

The components of this latent variable model are defined as follows:

- $P_i \sim \mathcal{N}(0, 1)$: The **static popularity** of contestant i , representing their baseline fan support independent of performance.
- $Q_{i,t}$: The **dynamic performance** signal $Q_{i,t}$ is constructed from the judges' evaluations. Let $J_{i,t}$ denote the total judges' score received by contestant i in week t . To ensure comparability across weeks, $Q_{i,t}$ is defined as the Z-score of $J_{i,t}$ within the set of contestants competing in week t , thereby capturing the relative quality of the contestant's performance in that episode.
- $\mu \sim \mathcal{N}(0.5, 1)$: The **influence weight** (parameters to be estimated) that determines the relative importance of popularity versus performance in driving votes.
- $\epsilon_{i,t} \sim \mathcal{N}(0, \sigma^2)$: A **stochastic noise term** representing unobserved idiosyncratic factors, such as "sympathy votes" after a poor performance or unexpected viral moments.

The latent scores are then mapped to the observable **vote share** $U_{i,t}$ through a Softmax function, ensuring the total share sums to unity:

$$U_{i,t} = \frac{\exp(A_{i,t})}{\sum_{j \in S_t} \exp(A_{j,t})} \quad (2)$$

where S_t is the set of contestants competing in week t .

Stochastic Approximation of Latent Voting Distributions:

The model characterizes the elimination event E_t by constructing a **probabilistic likelihood function** that mirrors the specific rules of each season. For each week t , we define the likelihood of contestant k being eliminated as a Softmax-based probability:

$$L(\theta | E) = \prod_{t=1}^T \frac{\exp(\delta \cdot S_{k,t})}{\sum_{j \in S_t} \exp(\delta \cdot S_{j,t})} \quad (3)$$

Here, θ is introduced as an intermediate variable, defined as

$$\theta = \{P_1, P_2, \dots, P_N, \mu\},$$

and E denotes the collection of all observed elimination events throughout the competition. $S_{i,t}$ represents the composite elimination score of contestant i at week t and δ serves as a sensitivity scaling parameter. The magnitude $|\delta|$ determines the discriminative power of the model. After iteratively tuning the magnitude of δ , we set $|\delta| = 3.0$. The definition of $S_{i,t}$ and The sign of δ vary according to the competition rules:

- **Rank-based Rule (Seasons 1–2, 28–34)**: The total score is the sum of ranks from judges and fans. Larger values indicate poorer performance, thus $\delta > 0$:

$$S_{i,t} = R(J_{i,t}) + R(U_{i,t}) \quad (4)$$

where $R(\cdot)$ denotes the descending rank function.

- **Percentage-based Rule (Seasons 3–27):** The score is a weighted sum of percentage shares. Smaller values indicate higher elimination risk, thus $\delta < 0$:

$$S_{i,t} = \frac{J_{i,t}}{\sum_{j \in S_t} J_{j,t}} + U_{i,t} \quad (5)$$

As the posterior distribution resulting from these non-linear elimination rules is **analytically intractable**, we employ **Markov Chain Monte Carlo (MCMC)**[3] sampling. By utilizing the Metropolis-Hastings algorithm, we generate a sequence of samples to approximate the high-dimensional parameter space of latent fan votes.

Result and Limitation Analysis:

Empirical testing revealed that Model I achieved an accuracy of only 62.09% in reproducing historical eliminations. Notably, it failed significantly when dealing with “low-score, high-popularity” contestants (e.g., Bobby Bones), often predicting a high elimination probability for contestants who actually won. We identified three inherent drawbacks:

- **Prior Suppression:** The Gaussian prior $P_i \sim \mathcal{N}(0, 1)$ causes “mean regression”, which systematically underestimates extreme popularity levels.
- **Soft Constraints:** The probabilistic likelihood allows for a non-zero probability of logically impossible outcomes, failing to strictly honor the known historical results.
- **Computational Inefficiency:** High-dimensional sampling requires excessive iterations ($> 50,000$), limiting its scalability.

Given that eliminations are deterministic historical facts, we concluded that a shift from “stochastic distribution fitting” to “constrained optimization” was necessary to accurately capture the voting dynamics.

3.2 Model II: Bayesian MAP Estimation via Constrained Optimization

3.2.1 Optimization Principle

To overcome the limitations of Model I, we reformulate the inference as a **Bayesian Maximum A Posteriori (MAP)** estimation problem, solved via **constrained Total Variation (TV) minimization**. In this framework, we treat latent popularity $P_{i,t}$ as a parameter set governed by a Laplace-type prior, which mathematically encodes the Parsimony Principle: popularity is inherently stable and evolves through sparse, incremental shifts.

The solver navigates this feasible region by invoking the **Parsimony Principle**, which assumes that a contestant’s latent popularity is inherently stable and changes only incrementally over time. Specifically, we impose a **Total Variation (TV) constraint**, assuming that fluctuations between adjacent weeks are smooth rather than erratic. In conclusion, Model II overcomes mean-reversion by allowing $P_{i,t}$ to vary dynamically and enforcing extreme values through hard elimination constraints when necessary.

3.2.2 Formal Redefinitions

- **Fan Voting Index**

$$A_{i,t} = (1 - \lambda)P_{i,t} + \lambda Q_{i,t} \quad (6)$$

where $\lambda \in [0, 1]$ represents the weight of performance relative to popularity. Unlike Model I, where μ is a random variable, in Model II we treat λ as a deterministic parameter optimized through grid search.

- **Vote Share (Using Softmax):**

$$U_{i,t} = \frac{\exp(A_{i,t})}{\sum_{j \in S_t} \exp(A_{j,t})} \quad (7)$$

- **Composite Scoring Rules ($S_{i,t}$):** The scoring follows the same rule of the Model I ($Rule_t$):

$$S_{i,t} = \begin{cases} Rank(J_{i,t}) + Rank(U_{i,t}) & \text{Rank-based (Lower score is better)} \\ \frac{J_{i,t}}{\sum J_{j,t}} + U_{i,t} & \text{Percentage-based (Higher score is better)} \end{cases} \quad (8)$$

3.2.3 Solution Procedure

1. **Dynamic Trajectory via Constrained Optimization:** We solve for the smoothest popularity trajectory by minimizing the Total Variation (TV) under hard constraints:

$$\min_{\{P_{i,t}\}, \lambda} \mathcal{L}_{TV} = \sum_i \sum_{t=2}^{T_i} |P_{i,t} - P_{i,t-1}| \quad (9)$$

$$\text{s.t. } S_{j,t} > S_{k,t} + \epsilon, \quad \forall j \in \text{Survivors}, k \in \text{Eliminated} \quad (10)$$

The TV objective encourages piecewise constant popularity, reflecting that fan bases are generally stable but can shift due to specific events. The safety margin ϵ ensures numerical stability at the decision boundaries.

2. **Static Popularity Extraction:** To obtain the inherent popularity P_i , we extract the time-average of the optimized dynamic trajectory:

$$\hat{P}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} P_{i,t} \quad (11)$$

where T_i denotes the total number of weeks that contestant i participates in the competition.

According to the **Minimum Variance Unbiased Estimator (MVUE)** theorem, $\hat{P}_i$ provides the most reliable unbiased estimate of a contestant's fundamental popularity.

3. **Computational Implementation** To recover the latent parameters, we implement a *two-stage computational pipeline* based on constrained optimization. Initially, judges' raw scores are transformed into dimensionless *Z-scores* to ensure data compatibility. The core estimation is then formulated as a *Grid Search* over the performance weight $\lambda \in [0, 1]$, where for

each candidate value, we solve a *Constrained Linear Programming* problem. This solver minimizes the *Total Variation* of popularity trajectories to satisfy all historical elimination boundaries as hard constraints.

Once the optimal λ^* is identified, we extract the static inherent popularity $\hat{P}_i$ through a *time-series aggregation* based on the *MVUE theorem*. These estimates are finally used to reconstruct the finalized fan vote shares $U_{i,t}$ via a *Softmax transformation*. The detailed execution logic is illustrated in the following workflow (Figure 3).

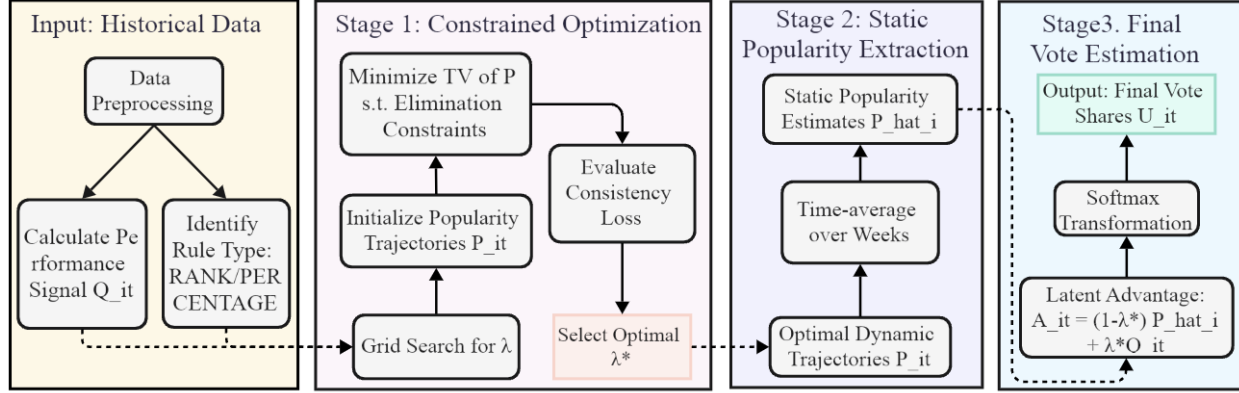


Figure 3: Model II Computational Workflow

3.3 Model Results and Uncertainty Analysis

Estimation Performance and Consistency

The validity of our estimates is evaluated by the model's capability to reproduce historical elimination outcomes. We define the *Consistency Accuracy* (C) as follows:

$$C = \frac{1}{T} \sum_{t=1}^T \mathbb{I}[E_{pred,t} = E_{actual,t}] \quad (12)$$

where T denotes the total number of elimination rounds, and $\mathbb{I}[\cdot]$ is the indicator function.

By conducting a grid search over the parameter space, our optimized Model II identified the *optimal performance weight* $\lambda^* = 0.4533$, suggesting that approximately 45.3% of the fan voting influence is driven by weekly dance performance, while the remaining 54.7% is attributed to inherent popularity.

With this optimized weight, the model achieved an impressive **99.14% constraint satisfaction rate** within the solver, successfully reconciling nearly all controversial historical cases (such as the “Bobby Bones” phenomenon) by identifying feasible popularity trajectories that align with deterministic elimination outcomes.

Estimation Certainty

To quantify the uncertainty of our point estimates, we performed a *sensitivity analysis* by introducing stochastic perturbations $\epsilon \sim \mathcal{N}(0, \sigma^2)$ to the optimal solution $U_{i,t}^*$. The estimation

certainty is characterized by the width of the 95% credible interval:

$$W_{i,t} = CI_{upper} - CI_{lower} \quad (13)$$

A smaller interval width (W) indicates higher certainty in the inferred vote share. In our model, this higher certainty typically occurs for contestants who were “near the bubble” (at high risk of elimination). Their vote shares are more tightly constrained by the observed survival/elimination outcomes, leaving less “slack” for variation in the feasible region. The distribution of estimation certainty across contestants and weeks in season 27 is visualized in Figure 4.

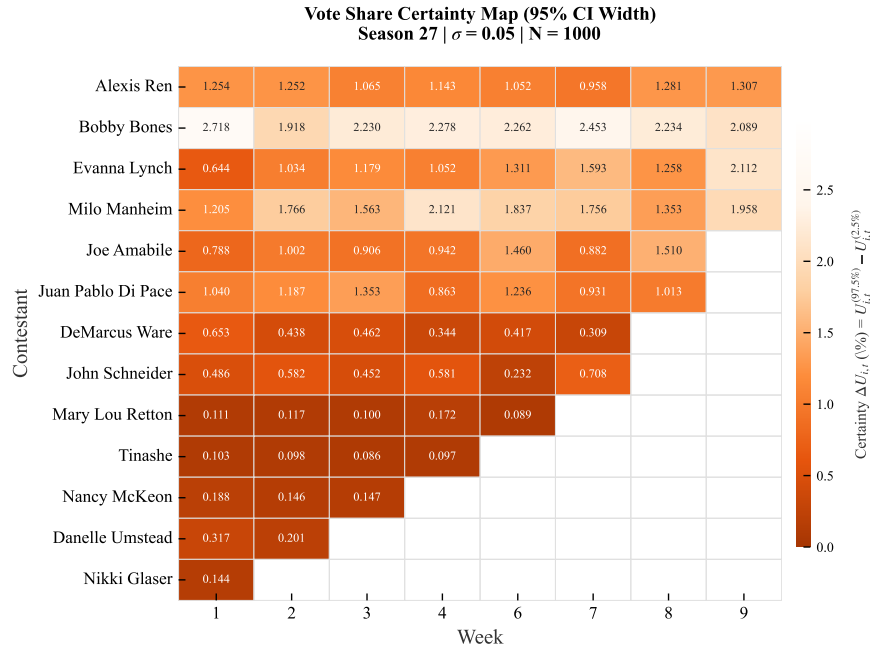


Figure 4: Heatmap of Estimation Certainty Across Contestants and Weeks (Season 27)

In the figure, the deeper the colour indicates a narrower range for W , thereby signifying a higher degree of certainty in the voting data estimation. As can be seen from the figure, our model demonstrated a high degree of certainty during the Season 27 validation.

4 Task 2: How Vote combination and Judge Saves Shape Competition Outcomes

4.1 Part 1: Rank-Based vs. Percentage-Based Methods Across Seasons

Problem Analysis

This section aims to compare whether the two voting combination and elimination rules employed in the show (the rank-based combination method and the percentage-based combination method) yield different elimination outcomes in actual competitions. It further analyses which rule is more inclined to reflect fan voting preferences overall.

To this end, while maintaining the season's data unchanged, we applied both sets of rules concurrently to each valid elimination week. By comparing the differences in the eliminated contestants, we conducted a quantitative assessment of the systemic bias inherent in the rules.

Definition of Match Result Differences Indicators

Suppose the final eliminated contestant in a given elimination week is defined as follows under the ranking system and percentage system respectively:

$$E_t^{\text{rank}}, \quad E_t^{\text{pct}}, \quad (14)$$

where $t = 1, \dots, T$ denotes the effective elimination weeks.

First, we define the *inconsistency rate* D to characterise the degree of divergence between two rules in their elimination outcomes:

$$D = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(E_t^{\text{rank}} \neq E_t^{\text{pct}}), \quad (15)$$

Here, $\mathbb{I}(\cdot)$ denotes the indicator function. It takes the value 1 when both rules eliminate different contestants in the same week, and 0 otherwise.

To measure the degree of alignment between the final elimination results and the preferences of judges or fans, we consider $J_t^{\min}$ and $U_t^{\min}$ as the contestants who received the lowest judges' score proportion and lowest fan voting rate, respectively. To quantify these relationships, we define:

- **Judge alignment Rate** C_J : The proportion of eliminated contestants whose scores align with the lowest judge's score for that week.

$$C_J = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(E_t = J_t^{\min}). \quad (16)$$

- **Fan Consistency Rate** C_U : The proportion of eliminated contestants matching the contestant with the lowest fan voting support rate that week.

$$C_U = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(E_t = U_t^{\min}). \quad (17)$$

These indicators are evaluated independently under both the Rank-based and Percentage-based rules. The resulting consistency levels are then compared across both the season level and the full-sample level.

Results Analysis and Conclusions

Based on the fan vote counts inferred from Task 1, we recalculated elimination outcomes for 34 seasons and 335 valid elimination weeks using both ranking and percentage systems. We have obtained a distribution chart illustrating the divergences in elimination outcomes across different seasons, as shown in the figure5.

Results indicate the inconsistency rate differs across seasons. By calculation an average **inconsistency rate** of approximately $D \approx 22\%$, meaning roughly one in five elimination weeks yielded different eliminated contestants under the two systems. This demonstrates that the chosen combination method does indeed exert a substantive influence on competition outcomes.

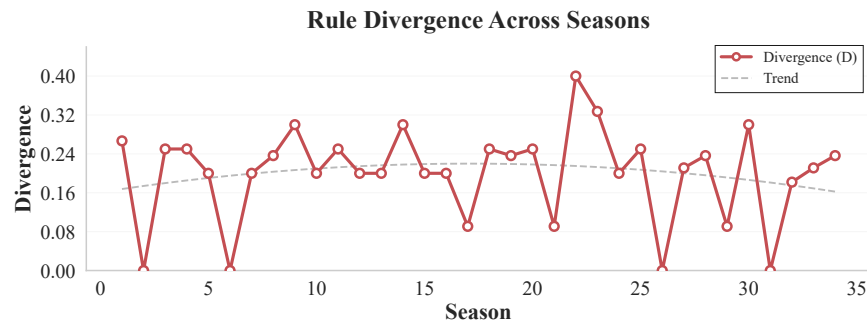


Figure 5: Elimination Outcomes Divergences Across Seasons

Further comparison of **judge alignment rates** versus **fan alignment rates** reveals that, at the season level, **the ranking system yields a marginally higher judge alignment rate than the percentage system (0.546 vs. 0.496)**. This indicates that under the ranking system, the final elimination outcomes generally align more closely with the judges' scoring order. Figure 6 illustrates the proportion of eliminated contestants who were the judges' lowest-ranked contestants under both systems across different seasons. Neither system demonstrated absolute superiority across all seasons.

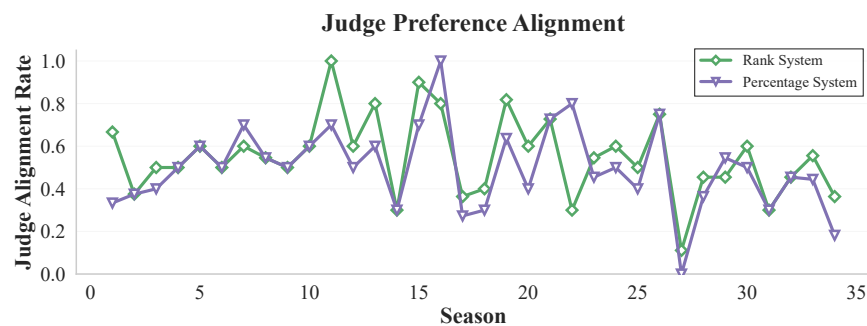


Figure 6: Judge Alignment Rate Across Seasons

By contrast, the percentage system demonstrates a more pronounced advantage in reflecting fan preferences. **Its fan alignment rate is markedly higher than that of the ranking system (0.627 vs. 0.460)**, indicating that under the percentage system, the final elimination outcome more frequently aligns with the contestant receiving the lowest support in fan votes. The results shown in Figure 7 reveal that for most seasons, the fan alignment rate corresponding to the percentage system (orange) consistently exceeds that of the ranking system (blue).

In summary, the two voting combination and elimination rules exhibit differences in their outcomes. The percentage system generally facilitates a stronger reflection of fan voting preferences in the final elimination results, whereas the ranking system aligns more closely with the judges' scoring order.

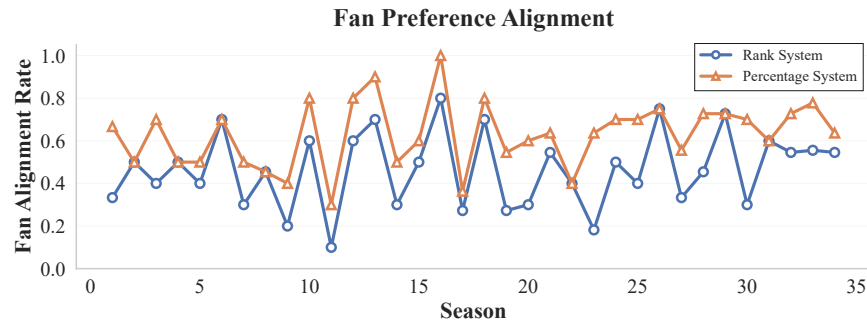


Figure 7: Fan Alignment Rate Across Seasons

4.2 Part 2: Controversial Cases Under Judge–Fan Disagreement

Problem Analysis

This issue centres on typical cases where judges’ scores and fan votes diverge significantly. It compares elimination outcomes under different judge-score versus fan-vote combination systems, further analysing how competition results change after introducing a new system. The new system entails: selecting the two contestants with the lowest combined rankings based on judges’ scores and fan votes, whereupon the judges vote to determine their final elimination.

For the first two existing combination methods (Rank-based and Percentage-based), we directly reconstruct the weekly elimination results based on the model established in Task 1. For the newly introduced system, assuming the judges’ choices in the final voting stage align with their original scoring rankings, we construct **Model III** to simulate the complete elimination process (see Figure8), thereby determining the contestants’ final elimination status.

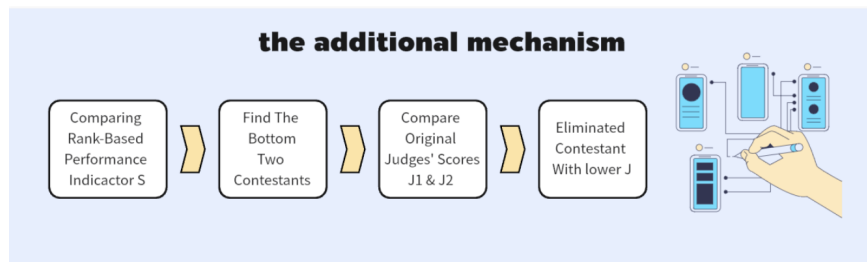


Figure 8: Elimination Process of the Additional System

Results Analysis

We compared the earliest elimination weeks for four contestants exhibiting pronounced “judges–fans divergence” across three competition formats, alongside their inferred final rankings (where 1 denotes the champion, with higher values indicating lower positions). The aggregated results are presented in Table1, where RANK means Rank-based system, PCT means Percentage-based system, BOT2 is our newly introduced system of the “Bottom Two” rule.

Table 1: Comparison of Elimination and Ranking Results Across Different Rules

Season	Contestant	Actual Result	Elim. Week (RANK)	Elim. Week (PCT)	Elim. Week (BOT2)	Rank (RANK)	Rank (PCT)	Rank (BOT2)
S2	Jerry Rice	Runner-up	W8	W7	W6	2	3	4
S4	Billy Ray	5th Place	<i>Not Elim.</i>	W8	W8	1	5	5
S11	Bristol Palin	3rd Place	W5	W10	W7	8	3	6
S27	Bobby Bones	Champion	<i>Not Elim.</i>	<i>Not Elim.</i>	<i>Not Elim.</i>	1	1	1

Jerry Rice performed best under the pure Rank system (which also served as the actual competition format for Season 2, where he ultimately placed second), dropped to third under the Percentage system, and became the lowest-ranked of the four under the new system.

Billy Ray Cyrus secured fifth place under both the Percentage system (the actual competition format) and the new system, yet was projected as the champion under the Rank system.

Bristol Palin ranked third under the percentage system (the actual competition format), dropped significantly to eighth under the Rank system, and placed seventh under the new system.

Bobby Bones maintained the champion position consistently across all three formats.

Comparing contestant rankings across different systems reveals significant discrepancies in extreme situations where judges' scores and fan votes diverge markedly. **The Rank and Percentage systems each exhibit distinct tendencies towards eliminating highly popular contestants.** However, after the introduction of judges' final decisions on the bottom two, **the new system enables earlier elimination** of contestants who consistently receive low judges' scores yet remain in the competition due to fan support. This enhances the professionalism of the final outcome to a certain extent.

4.3 Part 3: Selecting a Robust Voting Rule: Mitigating Extreme Judge–Fan Mismatches

Based on the analyses in the first two parts of Task 2, we observe that the three evaluation systems yield largely consistent rankings for most contestants. However, notable discrepancies arise in **extreme cases** where fan voting exhibits strong bias, particularly for contestants with high popularity but relatively weak dance performance. Such situations may undermine the professional integrity of a dance competition and reduce fairness in performance-based evaluation.

Combining the quantitative results from part 1, it is evident that *the percentage-based combination system assigns a greater statistical weight to fan votes*, thereby amplifying popularity effects in the composite score S . In contrast, *the rank-based system compresses extreme fan support and limits the influence of disproportionately high popularity*, allowing judges' scores to play a more stable role in determining competitive outcomes. From the perspective of maintaining professionalism and competitive fairness, **the rank-based combination** is therefore preferred.

Furthermore, *the additional system introduced in Part 2 demonstrates strong robustness in mitigating extreme cases*. By allowing judges to determine elimination outcomes from the bottom two contestants identified by the composite ranking, this approach preserves audience engagement while reinforcing expert judgment. Consequently, we recommend adopting a **rank-based combination framework**, including the **additional approach** of judges choosing from the bottom two contestants.

5 Task 3: Do Professional Dancers and Contestant Characteristics Matter?

5.1 Motivation and Modeling Philosophy

After inferring latent fan voting behavior in Task 1, we investigate how contestant characteristics (such as age and industry) and professional dancers influence competition outcomes, and whether these factors affect judges' scores and fan votes differently.

This is a *hierarchical* modeling problem. The same professional dancer may appear across multiple seasons and be paired with different celebrities, inducing *repeated, non-independent observations*. In addition, the observed variation in outcomes arises from both contestant-specific attributes and persistent dancer-specific influences, which must be explicitly separated to avoid confounding.

To address these issues while incorporating both *continuous* and *categorical* covariates, we adopt a **Model IV: Linear Mixed Effects Model (LMM)** [4] on **Z-score standardized variables**, treating contestant characteristics as fixed effects and professional dancers as random effects. This framework integrates the Maximum Likelihood Estimation (MLE) of fixed effects—capturing the universal impact of contestant characteristics—with the prediction of random effects for each professional dancer, effectively isolating dancer-specific influence from the broader scoring patterns.

5.2 Model IV's Dual-Model Framework: Judges Versus Fans

A central hypothesis of this study is that judges and fans evaluate contestants according to different criteria. Moreover, they collectively determined the outcome of the match. Judges are expected to emphasize technical execution and improvement, while fans may be more responsive to personality, narrative appeal, or pre-existing popularity. To test this hypothesis, we construct two parallel mixed-effects models.

5.2.1 Model A: The Judge Model

The first model takes standardized judges' scores as the dependent variable:

$$\hat{J}_{i,t} = \beta_0 + \beta_{age} \cdot \text{Age}_i + \sum \beta_{area_i} \cdot \text{Area}_i + \sum \beta_{ind_i} \cdot \text{Industry}_i + u_{pro} + \epsilon_{i,t} \quad (18)$$

Unlike the original $J_{i,t}$, which represent the total judges' score for contestant i in week t , $\hat{J}_{i,t}$ is defined as the *Z-score standardized judges' score*. It is calculated by subtracting the seasonal mean and dividing by the standard deviation. This allows for comparison of judges' scoring variability across different seasons.

Here, the *fixed effects* correspond to observable contestant characteristics, with coefficients β_{age} for age, β_{area} for geographic region, and β_{ind} for industry; these capture the general patterns of how such attributes influence judges' scores. The *random effect* u_{pro} represents the contribution of the professional dancer (pro-dancer synergy effect). A large positive u_{pro} indicates that a professional dancer consistently elevates judges' scores, regardless of the contestant's background, reflecting superior instruction, choreography design, or the ability to highlight strengths and conceal weaknesses.

5.2.2 Model B: The Fan Model

The second model focuses on fan voting behavior. Using the inferred vote share from Task 1, we apply a logit transformation to obtain a approximately normal response variable:

$$\hat{A}_{i,t} = \alpha_0 + \alpha_{age} \cdot \text{Age}_i + \sum \alpha_{area_i} \cdot \text{Area}_i + \sum \alpha_{ind_i} \cdot \text{Industry}_i + \alpha_{judge} \cdot J_{i,t} + v_{pro} + \epsilon_{i,t}. \quad (19)$$

Similarly, $\hat{A}_{i,t}$ denote the Z-score standardized fan vote index for contestant i in week t .

Here, the *fixed effects* correspond to observable contestant characteristics, with coefficients α_{age} for age, α_{area} for geographic region, α_{ind} for industry, and α_{judge} for judges' scores; these capture how each attribute influences fan voting behavior. Including judges' scores as a control allows us to isolate fan preferences that are not purely driven by technical performance. The *random effect* v_{pro} represents the contribution of the professional dancer(pro-dancer synergy effect), reflecting their popularity or fan appeal, which can affect votes independently of judges' evaluations.

5.3 Results

By utilizing Maximum Likelihood Estimation (MLE), we derived the coefficient estimates for fixed effects, which quantify the marginal impact of factors such as age and industry. This approach allows for a direct magnitude comparison of these estimates across the Judge and Fan models, revealing whether fan vote and judges' scores are structurally aligned to the contestant-specific attributes.

To isolate the pro-dancer synergy Effect, we extracted the random intercepts u_{pro} and v_{pro} using Best Linear Unbiased Prediction (BLUP) [5]. These estimates represent the systematic deviation from the predicted score assigned to each professional partner.

- **judges' scores effect on fan vote index:**

We generated a regression plot (Figure 9) illustrating the effect of standardized judges' scores ($\hat{J}_{i,t}$) on the fan voting index ($\hat{A}_{i,t}$) to evaluate their relationship.

The solid blue line represents a regression line with a positive slope, indicating a statistically positive correlation between judges' scores and fan voting behavior.

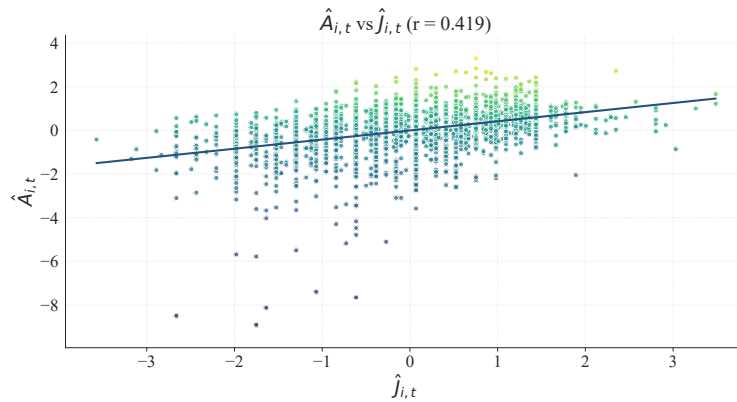


Figure 9: The Effect of Judges' Scores on Fan Vote Index

- **age effects:**

By generating regression plots (Figure 10) of Age against standardized judges' scores ($\hat{J}_{i,t}$) and fan voting indices ($\hat{A}_{i,t}$), we can observe how age impact the judges' scores and fan votes: Judges' scores $\hat{J}$: The plot shows a clear downward-sloping regression line with a correlation

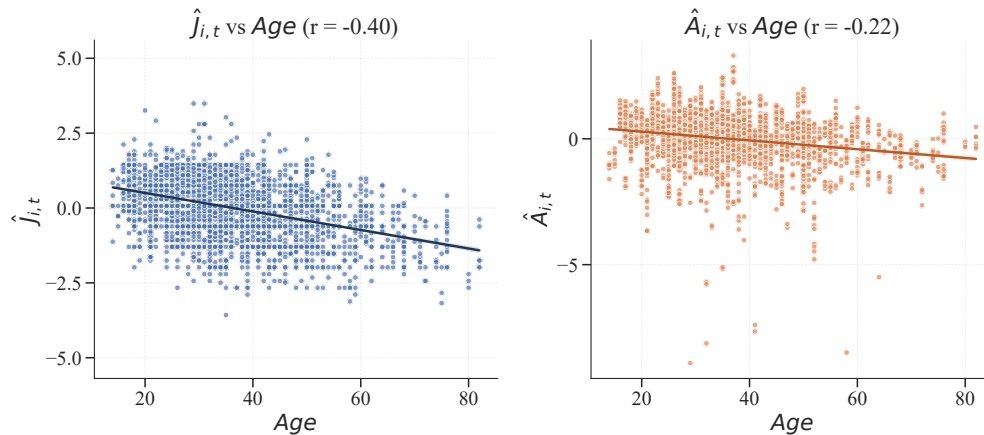


Figure 10: The Effect of Contestant Age on Judges' Scores and Fan Vote Index

coefficient of $r = -0.40$.

Fan Voting Index $\hat{A}$: the slope is noticeably flatter with a lower correlation of $r = -0.22$

As contestant age increases, **their standardized judges' scores tend to decrease more sharply than their fan support.**

- **region effects:**

We estimate separate effects of different regions on judges' scores and fan votes, with all coefficients interpreted relative to a pooled reference group consisting of the remaining regions that are not individually reported due to limited sample sizes. As shown in Figure 11:

Specifically, **Canada** exhibits a relatively larger positive effect on judges' scores, indicating that judges tend to favor participants from this region. In contrast, **Australia** displays a significant negative association with judges' evaluations, while the effects for England and Europe remain relatively marginal.

Regarding fan preferences, **England and Europe** show particularly strong promoting effects on fan voting outcomes, with their coefficient estimates reaching the highest levels among all groups.

Comparing the differential impacts on the two models, we find divergences in several regions. For instance, while **Australia** shows a negative impact on judges' scores, it receives positive support from fan votes. Similarly, England and Europe exert a neutral or mild effect on professional scores but *strongly enhance fan voting support*. These results highlight a clear discrepancy between judges' scores and fan votes across different regional identities.

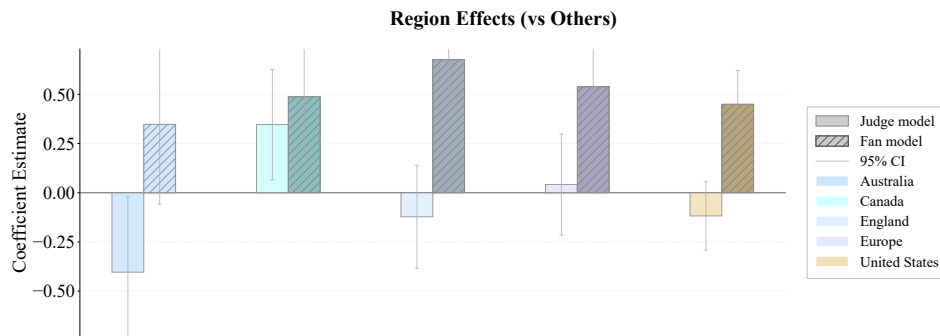


Figure 11: The Effect of Region on Judges' Scores and Fan Vote Index

- **occupation effects:**

We estimate separate coefficient effects of different occupations on judges' scores and fan votes, with all coefficients interpreted relative to a pooled baseline consisting of the remaining occupations that are not individually reported due to limited sample sizes. As shown in Figure 12, substantial heterogeneity across occupations can be observed.



Figure 12: The Effect of Occupation on Judges' Scores and Fan Vote Index

Specifically, **actors/models and singers** exhibit relatively larger effects on judges' scores: actors and singers contribute positively to judges' evaluations, whereas models display a negative association.

In contrast, the occupation of model shows a pronounced suppressing effect on fan votes, while **TV personalities and comedians** significantly promote fan voting outcomes.

Comparing the differential impacts on the two models, we find that actors and singers, *although improving judges' scores, tend to reduce fan voting support*. Conversely, *TV personalities strongly enhance fan votes but exert a mild negative effect on judges' scores*. These results highlight a clear divergence in how occupational identities influence expert evaluation and public preference.

- **pro-dancer synergy effects:**

We further examine the synergy effects of professional dancers on the scores awarded by judges. Preliminary analysis indicates that the identity of the professional dancer has a

negligible impact on fan voting patterns. Consequently, we focus our analysis exclusively on the pro-dancer synergy effects regarding judges' evaluations, as illustrated in Figure 13.

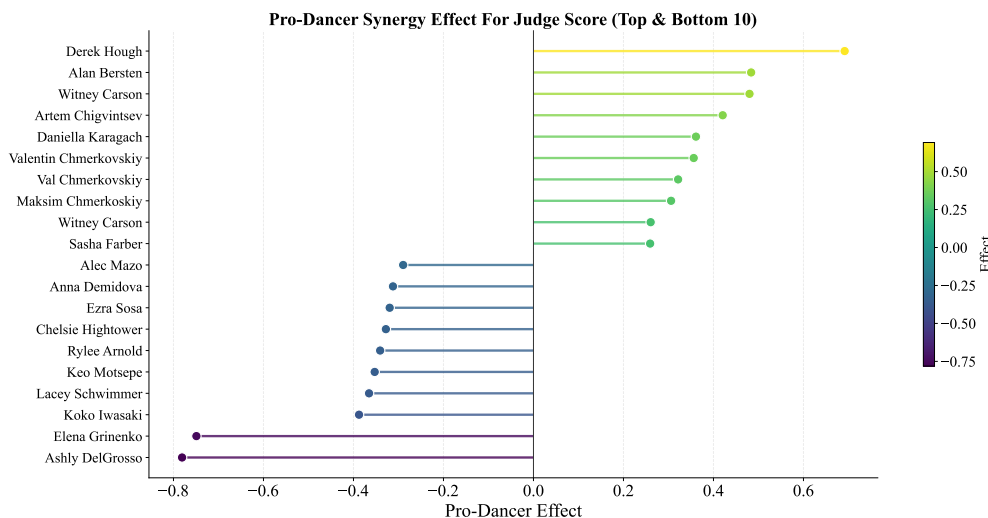


Figure 13: The Pro-dancer Synergy Effects on Judges' Scores

As shown in the visualization, there is a pronounced disparity in how different professional dancers influence the scoring outcomes. The *random effect* v_{pro} reveals that specific professional dancers introduce a significant shift in the model's intercept, reflecting their unique contribution to the judges' scores. Specifically, **Derek Hough** stands out with the most substantial positive synergy effect, significantly elevating the judges' scores compared to other professionals.

Conversely, certain professional dancers appear to have a suppressing effect on the baseline score. For instance, **Ashly DelGrosso** and **Elena Grinenko** exhibit the most significant negative offsets, with their random effect values falling below -0.7 . These results confirm the necessity of including the random effect term v_{pro} , as the inherent influence of specific professional dancers plays a critical role in shifting the final scoring outcomes independently of other variables.

Summary

This analysis systematically examines the impact of professional dancers, as well as observable contestant characteristics (such as age and industry background), on judges' evaluations and contestants' scores. The results indicate that these factors do not affect judges and fans in the same way. Professional dancers influence competition outcomes through both instructional effects and popularity-driven mechanisms.

By explicitly modeling these multiple layers within a mixed-effects framework, we provide a coherent explanation for the observed discrepancies between judges' scores and fan voting behavior.

6 Task4:Selecting a Robust Voting Rule: Mitigating Extreme Judge–Fan Mismatches

Design Rationale

Analysis of the preceding two questions reveals that extreme discrepancies between judges' scores and fan votes predominantly stem from the undue dominance of fan voting in final elimination decisions. Contestants' popularity, fan base size, and voting fervour have to some extent obscured the professional performance reflected in judges' scores, thereby undermining the professionalism and consistency of competition outcomes.

However, as a televised competitive programme, entirely eliminating fan voting's influence is equally inadvisable. Fan engagement is a crucial factor in boosting programme ratings and audience retention. Consequently, voting outcomes should to some extent reflect viewer preferences, yet must not permit contestants with demonstrably inadequate professional performance to persistently evade elimination through reliance on popularity.

Design Principles

Based on the aforementioned trade-offs, we propose a *dynamic weighted voting synthesis system*(**Model V**): the weighting ratio between judges' scores and fan votes in the overall results is adaptively adjusted according to the contestant's weekly judges' scoring level. Specifically, when a contestant's judges' scores are at a reasonable level or above, judges' scores and fan votes carry comparable weight in the overall score to preserve the significant influence of fan votes on the competition's outcome. Conversely, when a contestant's judges' scores are markedly low, the weight of judges' scores in the overall score is increased to mitigate excessive interference from popularity factors in professional elimination results.

Model Formulation

Under this system, the comprehensive performance score for contestant i in week t is defined as

$$S_{i,t} = \begin{cases} w(J) \cdot R(J_{i,t}) + (1 - w(J)) \cdot R(U_{i,t}), & \text{Rank-based rule,} \\ w(J) \cdot \frac{J_{i,t}}{\sum_{j \in S_t} J_{j,t}} + (1 - w(J)) \cdot U_{i,t}, & \text{Percentage-based rule.} \end{cases} \quad (20)$$

Here, $J = J_{i,t} / \sum_{j \in S_t} J_{j,t}$ refers to the proportion of the judge's score for contestant i relative to the total scores of all contestants that week. The term $w(J)$ denotes the dynamic weighting of the judge's score within the overall result, determined by the contestant's weekly performance level among judges. It is defined as follows:

$$w(J) = \Delta\omega \cdot \frac{1}{1 + e^{k(J-\theta)}} + \omega_0. \quad (21)$$

- θ : The pass threshold for judges' scores, used to distinguish between normal performance and significantly inadequate performance. set to $\theta = \frac{3}{5N}$, where N denotes the number of contestants participating that week, set to $N = 10$;
- k : Steepness parameter controlling the sensitivity of judge weighting to score changes, set to $k = 100$;
- $\Delta\omega$: Adjustable weighting range representing the maximum increment in judge weighting, set to $\Delta\omega = 0.3$;

- ω_0 : Minimum weight for judges' scores, ensuring fan votes remain part of the decision-making process, set to $\omega_0 = 0.50$.

Evaluation of the New System's Effectiveness

To assess the feasibility of this new system, we utilised the judges' scores and voting estimation model derived from Task 1 to statistically analyse the full season judges' scores alongside fan voting data. Based on this, we simulated 10,000 matches involving 10 contestants and three judges. Based on this virtual dataset, we generated scatter plots to compare ranking shifts between the two systems, where "rank" refers to the final elimination order.

To facilitate a rigorous comparison, let Rank_{New} and Rank_{Old} denote a contestant's rank under the proposed and baseline systems, respectively. Furthermore, let $\mathcal{S}_{\text{low}}$ be the set of contestants whose judge score proportion J falls below the threshold θ , where $N_{\text{low}} = |\mathcal{S}_{\text{low}}|$. We define the following comparative variables:

$$R_{\text{drop}} = \frac{1}{N_{\text{low}}}(\text{Rank}_{\text{New}} - \text{Rank}_{\text{Old}}) \quad (22)$$

A significantly positive or negative value indicates the promotion or demotion impact of new system on the rank of contestants whose judges' scores below a specified threshold.

$$M_{\text{rev}} = P(\text{Survivor} \mid J < \theta, J = \text{Bottom}) \quad (23)$$

This measures the probability that a contestant identified by judges as the worst performer reversed the elimination outcome due to large number of votes cast by fans.

$$\rho_{\text{safe}} = P(\text{Rank}_{\text{New}} = \text{Rank}_{\text{Old}} \mid J > \theta) \quad (24)$$

The function Corr determines whether rankings are consistent. The expression measures the probability that a contestant's ranking under the new system remains identical to that under the old system when judges' scores exceed a specified threshold.

Results Analysis

Through Python calculations, we obtain that:

- $R_{\text{drop}} = +1.88$, indicating that the new system causes contestants with low judges' scores to drop in overall ranking, thereby diminishing the advantage conferred by fan votes;
- $M_{\text{rev}} = 0.27$, lower than the Rank system (0.42), Percent system (0.37), and the "judges' decision on the bottom two eliminations" mechanism (0.32), indicating that under the new system, fan votes have reduced influence over altering the final elimination outcomes for low-scoring contestants, effectively lowering the probability of controversy;
- $\rho_{\text{safe}} > 0.95$ indicates that when contestants perform as expected, elimination outcomes remain highly consistent with the original true results. This demonstrates that the system maintains fairness while still encouraging fan voting participation.

To assess the impact of the new system on ranking changes, we plotted a scatter diagram of contestants' ranking shifts: the horizontal axis represents the proportion of judges' scores obtained by each contestant, while the vertical axis indicates the magnitude of their ranking change.

The chart14 indicates that when the proportion of low-scoring judges falls below 0.1, particularly below 0.06, contestants' rankings decline markedly (upper half of the chart). Conversely, when the proportion of low-scoring judges exceeds this threshold, ranking changes predominantly remain stable or improve. This demonstrates that the new system effectively suppresses the rankings of contestants receiving low scores from judges, achieving the intended outcome.

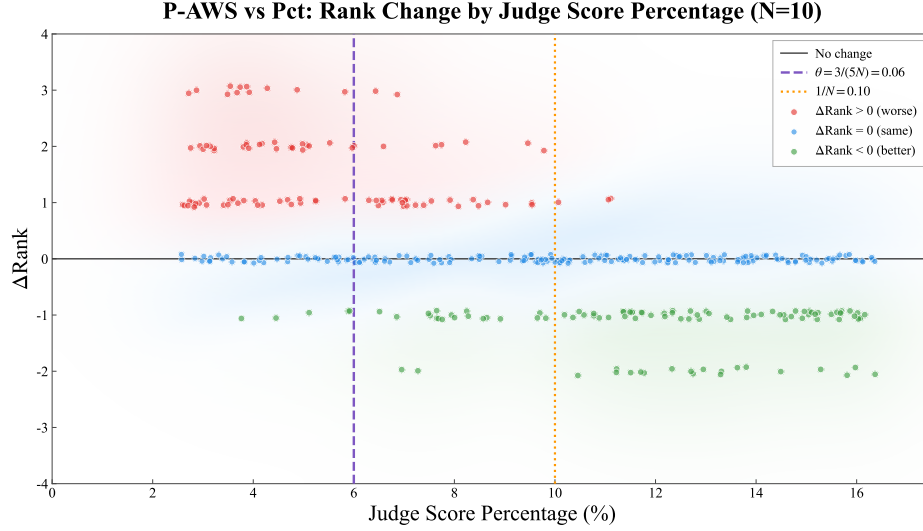


Figure 14: Scatter Diagram of Contestants' Ranking Shifts

7 Sensitivity Analysis

To evaluate the robustness of **Model V** proposed in Task 4, we conduct a stability analysis by perturbing its key parameters and examining the resulting variations in the evaluation metrics R_{drop} , M_{rev} , and ρ_{safe} .

We define $c = \theta \cdot N$ as an auxiliary intermediate quantity, introduced solely for stability analysis to control the combined influence of θ and N on the system behavior.

Specifically, we vary the parameters c , k and $\Delta\omega$, while keeping all other settings unchanged. Based on the computations described in Task 4, the results of the sensitivity analysis for the three metrics are presented in Figures 15–17.

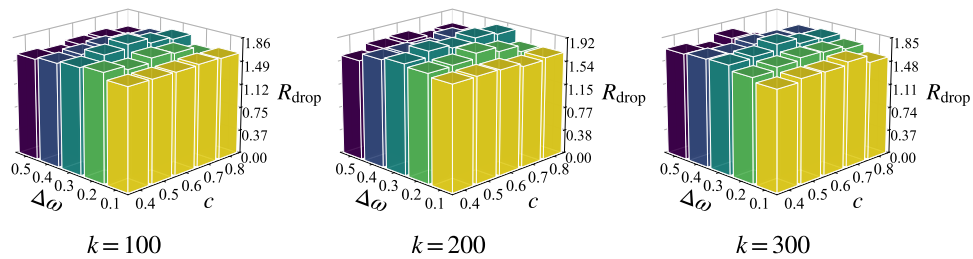
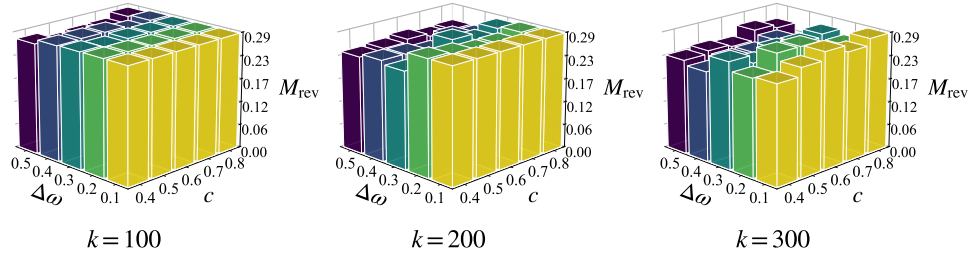
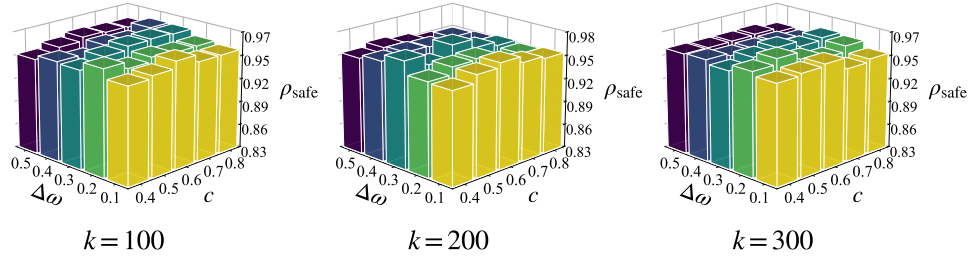


Figure 15: Sensitivity Analysis for R_{drop}

Figure 16: Sensitivity Analysis for M_{rev} Figure 17: Sensitivity Analysis for ρ_{safe}

The results indicate that all three indicators exhibit only mild variations under reasonable parameter perturbations, with no abrupt changes or trend reversals. This demonstrates that the performance of Model V is stable with respect to parameter variations, confirming the robustness and practical reliability of the proposed dynamic voting mechanism.

8 Model Evaluation

8.1 Strengths

- Inferring fan votes from the elimination list is a classic inverse problem, well-suited for a Bayesian latent variable model to recover unobservable votes.
- We transformed the high-dimensional Bayesian posterior sampling into a constrained convex optimization problem, achieving faster convergence and deterministic results.
- In the Linear Mixed Model (LMM), professional dancers were modeled as random effects ($u_{pro}, v_{pro} \sim N(0, \sigma^2)$), reducing parameters and enabling precise estimation of fixed effects like industry and age.
- The dynamic weighted voting system is cleverly designed and easy to implement, balancing judges' authority with fan engagement to maintain audience enthusiasm.

8.2 Weaknesses and Improvements

- The MAP-based constrained optimization model is highly accurate near elimination but less certain for safe contestants. It yields a single trajectory rather than a full posterior; approximated confidence intervals are less rigorous than full Bayesian inference.
- Task 2 analysis of Rank and Percentage scoring focused on data rather than theoretical mechanism properties. Future work could analyze fan preferences from the theoretical perspective of the scoring systems.
- LMM's linear structure limits capturing non-linear factor effects. Future extensions could introduce non-linear terms or use a Generalized Additive Mixed Model (GAMM) to model complex relationships.

9 Conclusion

This study quantitatively examines the interplay between judges' scores and fan votes in Dancing with the Stars, highlighting fairness and robustness. By modeling fan voting as a constrained inverse problem, we accurately reconstruct latent vote support. Comparative analysis shows rank-based rules favor judges, percentage-based rules favor fans. Introducing a "bottom-two with judge save" system moderates extreme outcomes while preserving audience influence. Based on these findings, we propose a dynamic vote-weighting system that adjusts the influence of fan votes according to judges' scores. Overall, the framework offers a transparent method for inferring hidden votes and provides actionable insights for designing fairer, credible competition rules, with applicability to other voting-based contests.

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- [2] Leonid I Rudin, Stanley Osher, and Emad Fatemi. Nonlinear total variation based noise removal algorithms. *Physica D: nonlinear phenomena*, 60(1-4):259–268, 1992.
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- [4] Nan M Laird and James H Ware. Random-effects models for longitudinal data. *Biometrics*, pages 963–974, 1982.
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Memorandum

Dear Producers:

We wanted to share some insights on the current way judge scores and audience votes are combined, and why it may be worth a closer look. Our focus is on how the rule itself shapes the overall competition results and influences elimination patterns across the show.

Our analysis points out that judges focus on technical performance, while fans vote based on preference or popularity. Both of the rank-based and percentage-based combination system had cause controversy over contestants with demonstrably inadequate professional performance to persistently evade elimination through reliance on fan vote.

The introduction of an additional mechanism in which judges determine the elimination between the bottom two contestants has helped reduce such controversies. However, this approach may also weaken audience motivation by limiting the impact of fan voting.



The design challenge, then, is clear: keep fans engaged and feeling influential, but reduce the chance that low-scoring contestants survive long purely because of votes, while maintaining fairness and professional credibility.

Our proposed approach keeps both judges' scores and fan votes in play but introduces dynamic weighting mechanism that adjusts the aggregation based on judges' scores. The idea is simple: adjust the combination between judges' scores and fan votes so that audience participation remains meaningful, while automatically amplifying the influence of judges'

scores in extreme cases—specifically when a contestant receives very low professional evaluations but exceptionally high fan support.

This approach helps maintain a smoother competition rhythm, by preserving sufficient room for fan voting and encouraging audience participation, while simultaneously reducing the likelihood of extreme and controversial outcomes and safeguarding the fairness and professionalism of the competition.

Finally, we want to emphasize: this analysis is meant to inform rule decisions, not to prove who's right or wrong. We hope it serves as a helpful reference for future seasons and look forward to collaborating on fine-tuning the competition rules.

We sincerely appreciate your attention and hope that these recommendations will provide valuable insights to guide future seasons of Dancing with the Stars.

Yours sincerely,

Team # 2612288



Report on Use of AI

1. Google Gemini (Feb 1, 2026 version, Gemini-3)

Using Gemini for translation, generating poster background images, assisting with problem comprehension, and learning modeling methods.

Query1: *Detailed introduction to the competition problem and dataset content, without citation links.*

Output:

Here is a detailed breakdown of the 2026 MCM Problem C: Data With The Stars and the accompanying dataset, tailored to your interest in statistics and modeling.

1. Problem Overview

The core theme of the problem is "Reverse Engineering" hidden data. You are analyzing the US TV show Dancing with the Stars (DWTS).

- The System: Celebrities are paired with professional dancers. Each week, they receive two types of scores: 1. Judge Scores (Observed): Technical scores from 3-4 judges (usually 1-10). 2. Fan Votes (Unobserved/Latent): The public votes for their favorites.*
- The Conflict: These two scores are combined to eliminate one couple each week. However, the show has faced "controversies" where popular celebrities (like Jerry Rice or Bobby Bones) survived or even won despite having terrible technical scores from judges.*
- The Goal: The show producers want to understand how different scoring methods affect outcomes and fairness.*

2. Specific Tasks

You must write a 25-page report addressing four main areas:

Task A: Reconstruct the Latent Variable (Fan Votes)

This is the most statistical part of the problem.

- Fan voting data is a "closely guarded secret."*
- You must build a mathematical model to estimate the fan votes for past seasons.*
- Validation: Your estimated fan votes, when combined with the known judge scores, must result in the correct historical eliminations. (This is where your Likelihood function $\mathcal{L}(\{P_{i,t}\}|\text{Data})$ comes in).*
- You need to quantify the certainty/uncertainty of your estimates (perfect for a Bayesian approach).*

Task B: Compare Voting Methods

The show has used two different mathematical methods to combine scores:

- (a) *Rank-Based (Seasons 1-2, 28-34): Total = Rank(Judges) + Rank(Fans). Lowest sum is eliminated.*
- (b) *Percentage-Based (Seasons 3-27): Total = %(Total Judge Points)+%(Total Fan Votes). Lowest percentage is eliminated.*

You need to analyze if changing the method changes the history. For example, if Season 2 used "Percentages" instead of "Ranks," would Jerry Rice still have reached the finals?

Task C: Variable Analysis

Using your estimated fan votes, analyze: The Pro Factor (do certain professional partners bring in more fan votes?) and Demographics (how do Age, Industry, or Home Country affect performance?).

Task D: Policy Recommendation

Propose a "fairer" or "more exciting" voting system for the future and justify it to the producers.

3. Dataset Content (2026_MCM_Problem_C_Data.csv)

The provided CSV file contains historical data for Seasons 1 through 34.

Key Columns:

- *Celebrity Information: celebrity name, ballroom partner, celebrity industry (e.g., "Athlete"), homestate/country, age.*
- *Outcomes: season, results (e.g., "Winner"), placement.*
- *Scores (The Time-Series Data): weekX_judgeY_score. Note: usually 3 judges, sometimes 4.*

Data Quirks & Handling:

- *0 Scores: Means the contestant had already been eliminated.*
- *N/A Values: No 4th judge or show didn't run that many weeks.*
- *Decimals: Some scores are averages of multiple dances.*

4. Why this fits your "Bayesian Latent Variable" plan

The problem explicitly asks for Measures of Certainty regarding the unknown fan votes. A traditional optimization approach cannot provide uncertainty intervals. A Bayesian MCMC approach will generate a distribution of possible fan votes, allowing you to say: "We are 90% sure Jerry Rice had between 22% and 25% of the vote."

—

Query2:I want to learn about Bayesian Latent Variable Models in detail.

Output:

This is a comprehensive overview of Bayesian Latent Variable Models (BLVMs). As a Statistics student, you will appreciate how these models combine the structural flexibility of latent variables with the probabilistic rigor of Bayesian inference.

1. The Core Concept

A Latent Variable Model assumes that the observed data (X) is generated by some underlying, unobserved (latent) variables (Z). Examples: Intelligence $\rightarrow$ Test Scores; Market Volatility $\rightarrow$ Stock Prices. A Bayesian approach adds two critical features: Priors (treating parameters θ as random variables) and Full Uncertainty (estimating the posterior distribution).

2. Mathematical Framework

In a BLVM, we define a joint probability distribution over the observed data X , the latent variables Z , and the model parameters θ :

$$P(X, Z, \theta) = \underbrace{P(X|Z, \theta)}_{\text{Likelihood}} \cdot \underbrace{P(Z|\theta)}_{\text{Latent Prior}} \cdot \underbrace{P(\theta)}_{\text{Parameter Prior}}$$

The Goal: Posterior Inference. We want to compute the posterior distribution of the hidden structure given the data:

$$P(Z, \theta|X) = \frac{P(X|Z, \theta)P(Z|\theta)P(\theta)}{\int \int P(X|Z, \theta)P(Z|\theta)P(\theta) dZ d\theta}$$

3. Common Types of BLVMs

Model Type	Latent Variable (Z)	Usage
Bayesian Mixture Models	Discrete Categorical	Clustering
Bayesian Factor Analysis	Continuous Vector	Dimensionality Reduction
Latent Dirichlet Allocation	Discrete Distribution	Topic Modeling
Hidden Markov Models	Discrete Sequential	Time Series
Variational Autoencoders	Continuous Non-linear	Generative AI

4. How We Solve Them (Inference)

A. Markov Chain Monte Carlo (MCMC): We simulate samples from the posterior. Technique: Gibbs Sampling. Pros: Asymptotically exact. Cons: Slow on large datasets.

B. Variational Inference (VI): Turn inference into an optimization problem (minimize KL Divergence). Technique: Maximizing ELBO. Pros: Fast, scales to massive data. Cons: Approximate.

5. Why use BLVMs for your specific needs? (High-dimensional problems $p > n$)

- Handling Missing Data: Impute missing values automatically.
- Regularization: Priors act as automatic regularizers (e.g., "Slab and Spike" for feature selection).
- Hierarchical Sharing: Share statistical strength across groups.

—

Query3:Detailed introduction to LLMs and comparison with Random Forest.

Output:

1. The Fundamental Difference

- *Random Forest (RF): Ensemble Method (Bagging Decision Trees). Discriminative ($X \rightarrow Y$). Best for structured tabular data.*
- *Large Language Models (LLMs): Generative Models (Transformer). Generative (Predict next token). Best for unstructured sequential data.*

2. Direct Comparison Table

<i>Feature</i>	<i>Random Forest</i>	<i>Large Language Model</i>
<i>Data Type</i>	<i>Structured / Tabular</i>	<i>Unstructured (Text)</i>
<i>Primary Function</i>	<i>Prediction</i>	<i>Generation</i>
<i>Training Cost</i>	<i>Low (CPU)</i>	<i>Massive (GPU)</i>
<i>Inference Speed</i>	<i>Microseconds</i>	<i>Seconds</i>
<i>Interpretability</i>	<i>Moderate (Feature Importance)</i>	<i>Low</i>

Summary: Use Random Forest for CSV files/Quantitative Finance. Use LLMs for raw text/Reasoning.

—

Query4: Help me come up with some good visualization methods.

Output:

- 1. Visualizing the Model: Probabilistic Graphical Model (Plate Notation) to show Bayesian structure. Methodology Flowchart.*
- 2. Visualizing the "Hidden" Fan Votes: Time Series with Uncertainty Bands (Ribbon Plot) to show 95% Credible Intervals. Ridgeline Plot (Joyplot) for distributions.*
- 3. Comparing Voting Methods: Bump Chart (Rank Tracking) to show rank crossings. Scatter Plot with 45-Degree Line.*
- 4. Analyzing Controversy: Bivariate Scatter Plot (Judge vs. Fan). Quadrants identify "Superstars", "Fan Favorites", etc.*

Recommendation: Plate Notation, Ribbon Plot, and Bump Chart.

—

Query5: Plan to analyze which scoring mechanism favors fan votes more.

Output:

Scheme Part 1: Theoretical Analysis (The Variance Argument). Principle: In a weighted sum, the component with higher Variance dominates. Rank Method has fixed variance. Percentage Method variance depends on data. Since Fan Vote variance (polarized) > Judge Score variance (clustered), Percentage Method favors fans.

Scheme Part 2: Counterfactual Simulation. Re-run history. Calculate "Save Rate": Count how many times the Rank vs. Percentage method saved a contestant with the lowest judge score.

Scheme Part 3: Correlation & Regression Analysis. Regress Final Outcome on Judge Score and Fan Vote. Compare ratio of coefficients β_2/β_1 .

Scheme Part 4: The "Bones" Test. Check if Bobby Bones (Season 27 Winner via Percentage) would lose under the Rank Method. If yes, Percentage Method favors fans.

—

2. Github CoPilot (Feb 3, 2024 version)

Auto-completions for code used in preparing our models.